

Element L

Design Recommendations and Plans for Implementation:

Recommendations	Rationale	Implementation
Ensure that the grooves are the same width throughout on the Onshape model before CNCing	The grooves on our final product were not the same width throughout, so it left an effect where the board looked significantly less bright in the middle because the strands were farther apart. If the grooves had been the same width throughout the entire board, it would have provided a much more complete and smooth effect because the strands would have sat right next to each other.	Redesign the onshape model using splines and the dimensioning tool instead of arcs to ensure that the width remains the exact same throughout the entire groove.
Use one consistent pattern / method of staining	Varying the size and shape of the staining pattern led to discrepancies in the coloration of the wood, and made some spots darker than others. Using one consistent pattern ensures that the stain is evenly distributed across the wood, and prevents any splotches.	Use the exact same staining pattern throughout the entire board from the start to ensure it all looks the same. Additionally, make sure the entire board surface is sanded down evenly to provide an unvarying surface for staining.
Pre-drill holes into the acrylic sheets before screwing them into the board.	Not pre-drilling the holes led to the acrylic sheets cracking when the group tried to screw them into the board. This slowed down the construction process and led to wasted material. Pre-drilling the holes would eliminate this setback and allow for a cleaner/more efficient installation.	Use the drill press to pre-drill holes in the acrylic sheet that are slightly larger than the screws. The holes must be slightly larger so that the screw doesn't actually drill into the acrylic sheet, as this is what causes it to crack.